

Anwesa Mondal

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in LinkedIn

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🌐 Portfolio

🔗 LeetCode

Education

Indira Gandhi Delhi Technical University for Women — CGPA 9.15

[Aug '24 – Present]

B.Tech in Computer Science Engineering with specialization in Artificial Intelligence

- **Computer Science:** Database Management System, Object-Oriented Programming, Design and Analysis of Algorithms
- **Artificial Intelligence:** Introduction to Data Science, Foundation and Application of Artificial Intelligence
- **Mathematics:** Probability and Statistics, Linear Algebra, Vector Calculus, Advanced Calculus, Discrete Mathematics

Experience

Harvard Medical School — Research Intern [[Letter of Recommendation](#)]

[Jun '25 – Oct '25]

- Under the supervision of Dr. Xiangrui Zeng, Laboratory for Computational Neuroimaging

- Implemented unsupervised surface registration and correction of 3D MRI images using neural deformation field learning
- Achieved **98.7%** Dice score by reducing **self-intersection loss** through training label maps that controlled deformation

National Institute of Technology, Rourkela — Research Intern [[Completion certificate](#)]

[May '25 – Jul '25]

- Under the supervision of Dr. Souvik Das, Department of Industrial Design

- Built a hybrid **Grid-Tuned LSSVM** and **Ensemble Learning** model for data imputation with **80%** missing rates
- Benchmarked against baselines on real-world multi-sensor time series, achieving **RMSE 0.0168** and **MAE 0.014** overall

Skills and Technologies

Programming Languages: C++, Python, SQL

Skills: DSA, OOP, DBMS, OS, Git, GitHub, REST APIs, Problem Solving, Machine Learning, Deep Learning

Python libraries: NumPy, Pandas, Matplotlib, Seaborn, OpenCV, Scikit-Learn, TensorFlow, Keras, PyTorch

Publications

A Grid-Tuned LSSVM and Ensemble Learning Framework for Missing Data Imputation [[Paper](#)]

- Presented and published in the conference proceedings of the 5th Indian IEOM Conference 2025, VIT Vellore, India

Projects

AI-powered road safety and insurance system — Google Solution Challenge, 2025

[SafeDrive](#) 🔗

- Achieved real-time drowsiness detection based on EAR values with **85%** accuracy leveraging **OpenCV** and **MediaPipe**
- Developed jerk analysis using speed data from Geolocation API and insurance validation with **Google Gemini 2.5 Pro**
- Implemented an in-app chatbot by providing **Gemini 2.5** with engineered prompts that increased engagement by **30%**

Stock Price Prediction using Long Short-Term Memory

[GitHub](#) 🔗

- Standardized dataset and engineered **sliding-window** sequences using **20** historical values to train an LSTM model
- Trained an **LSTM** using Chinese historical stock data, achieving **0.11 MSE (0.33 RMSE)** on real stock price forecasts

Lightweight User-Space Scheduler with C++20 Coroutines

[Github](#) 🔗

- Developed a C++20 coroutine runtime with work stealing and thread-pool scheduling, achieving **138M yields/s**.
- Built an epoll-based asynchronous TCP runtime, sustaining **117 MB/s** throughput for **1000 concurrent clients**.

Achievements

- Selected for **Amazon ML Summer School 2026**, placing among the **top 2.2%** of over **134,000** applicants nationwide.
- Solved **750+** problems on **LeetCode** involving various algorithms and achieved a peak rating of **1802** (top **7%** globally)
- Achieved a max rating of **1432 (2 *)** on **CodeChef** and **1224 (Pupil)** on **Codeforces**. [[CodeChef](#) 🔗][[Codeforces](#) 🔗]
- Solved **1000+** problems overall, across various platforms like **GeeksForGeeks**, **Code360**, and **HackerRank**

Competitions

- **Winner** at **Industrial Ideathon 2025** among **400+** teams across India, organised by the Government of Delhi
- Secured **top 10** position and advanced to the finals from among **130+** teams in **Ignitathon 2025**, organised by RCCIIT
- Progressed to the finale (**top 17** among **200+**) of **SpaceCon**, Space-Tech Hackathon, organised by NSUT [[Certificates](#)]
- Ranked among the **top 15** teams from **300+** teams in **Vihaan 8.0**, National Level Hackathon organised by DTU
- Finished among the **top 20** teams at **Innov8 3.0**, AI & Innovation Challenge, organised by Eightfold.ai [[Certificates](#)]